SUMAN KUMARAJOTHI

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SUMMARY

Software Engineer with 5 years of experience in IoT and telecommunications, specializing in backend development and scalable distributed systems. Proficient in Java, Spring Boot, Apache Kafka, and MQTT, with expertise in microservices architecture and modernizing legacy systems. Experienced in building reliable, high-performance solutions and driving engineering excellence through technical leadership and cross-team collaboration.

EXPERIENCE

Dhyan Networks & Technologies, Chennai, India

Lead Engineer — *Apr 2026 - Present*

SM NexGen Backend | *Java, Spring Boot, Apache Kafka, MQTT, Project Reactor, PostgreSQL, Redis, Docker, Kubernetes, Grafana, Loki*

- Led backend development for *StreetMan NexGen*, coordinating implementation planning, technical reviews, release preparation, and cross-team integration to ensure predictable delivery.
- Improved Kafka-based service communication by refining consumer-group management and messaging configurations, enhancing reliability and reducing latency during dynamic topic changes.
- Optimized asynchronous *MQTT* response correlation by replacing iterative publisher-session matching with token-based lookups and *TTL*-based session caching, reducing lookup overhead and improving scalability.
- Enhanced cross-service observability by propagating request context and implementing structured logging with *MDC*, enabling request tracing across vendor services using device and message identifiers.

Senior Software Engineer — *Apr 2024 - Mar 2026*

SM NexGen Backend | *Java, Spring Boot, Apache Kafka, MQTT, Project Reactor*

- Redesigned legacy *CMS* southbound vendor components as Spring Boot microservices, introducing reusable adapter patterns that improved modularity and simplified integration of additional device platforms.
- Designed foundational vendor-adapter components, including a base adapter, API resolver, and shared messaging abstractions, standardizing device-specific API integration across services.
- Developed a reusable *MQTT*-based southbound adapter library using Spring Integration and Project Reactor for asynchronous response handling, supporting concurrent device communication and maintainable messaging workflows.
- Refactored vendor messaging components using a modular, callback-based architecture to eliminate cyclic dependencies, improving maintainability and extensibility.
- Developed vendor-specific event processors to handle performance statistics and device events, enabling seamless integration with the DMS framework for downstream processing and persistence.

Dhyan LMS | *PHP, MySQL, Linux*

- Migrated the legacy leave-management application from *CentOS 7 (PHP 5.6)* to *Rocky Linux 9 (PHP 7.4)*, resolving compatibility and production issues across application workflows, database synchronization, SMTP, and server operations to ensure application continuity and stability.

iS5 Communication | *Java, JSP, jQuery, XML, PostgreSQL, SNMP, CLI*

- Led the development of a dynamic device-integration framework that replaced code-intensive onboarding with schema-driven configuration, simplifying device-integration workflows.
- Redesigned tightly coupled inventory collection into an annotation-based processing framework, enabling inventory attributes to be incorporated through structured schema definitions.
- Developed a dynamic chassis-rendering framework that updates hardware representations based on collected inventory, accurately reflecting the insertion and removal of supported hot-swappable modules.

- Facilitated knowledge transfer on *RAPTOREye* and *NetMan* infrastructure for the client development team and mentored interns through proof-of-concept assignments.

Software Engineer *Nov 2021 - Mar 2024*

SM NexGen UI | *Angular 17, TypeScript, PrimeNG, REST APIs*

- Led a small team in migrating the legacy UI to Angular and PrimeNG using reusable, schema-driven components and REST APIs, improving maintainability and development efficiency through coordinated cross-team integration.
- Developed a reusable hierarchical tree-view component for navigating deployment regions and asset groups, enabling context-aware filtering of discovered assets.
- Extended the tree-view component with tenant switching, enabling administrators to seamlessly navigate multiple deployments within the current login context.
- Redesigned and implemented the Dimming Scheduler component using *FullCalendar.js*, enabling users to configure smart-lamp lighting schedules through an intuitive, responsive interface.

Microsoft Affirmed Networks | *Java, JSP, Perl, MySQL, YANG, Azure DevOps*

- Developed northbound REST APIs for automatic registration of Network Functions in *Acuitas*, enabling network-element discovery, updates, and deletion through existing EMS device-management functionality.
- Improved EMS server reliability in High Availability and Disaster Recovery (*HA-DR*) systems by resolving slave-node alarm-generation issues and developing simulations to investigate split-brain conditions triggered by recurring switchovers.
- Refactored EMS statistics-model generation and validation scripts to resolve YANG schema-processing failures caused by symbolic-link changes, addressing *MCC 14.0* release issues and *MCC 13.1* backward-compatibility requirements.
- Remediated security vulnerabilities and static-analysis findings identified through *CodeQL, ESLint*, and *Azure DevOps Component Governance*, improving code quality and supporting release stability.
- Received the *Customer Excellence Award* for resolving critical issues that unblocked release workflows.

CNMS - S&C | *Java, JSP, JavaScript, PostgreSQL*

- Improved CNMS application stability and reliability by resolving UI/UX inconsistencies and functional defects across Discovery Filters, Alarms, Events, and Real-Time Statistics.

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, TypeScript, PHP, SQL, Perl, Shell scripting
Backend & Architecture: Spring Boot, Java EE, Hibernate, Project Reactor, Microservices, Distributed Systems
Frontend: Angular, PrimeNG, jQuery, HTML5, CSS3
Messaging & Protocols: Apache Kafka, MQTT, REST APIs, SNMP, CLI, HTTP
Databases & Data Formats: PostgreSQL, MySQL, Redis, XML, YANG
Tools & Infrastructure: Docker, Kubernetes, Azure DevOps, VMware vSphere, Grafana
Testing & Code Quality: JUnit, Mockito, Postman

EDUCATION

Bachelor of Engineering, Mechanical Engineering | CGPA: 7.94 *Aug 2015 - May 2019*
Panimalar Institute of Technology, Chennai, India